

John Benjamins Publishing Company



This is a contribution from *Scientific Study of Literature 4:1*

© 2014. John Benjamins Publishing Company

This electronic file may not be altered in any way.

The author(s) of this article is/are permitted to use this PDF file to generate printed copies to be used by way of offprints, for their personal use only.

Permission is granted by the publishers to post this file on a closed server which is accessible only to members (students and faculty) of the author's/s' institute. It is not permitted to post this PDF on the internet, or to share it on sites such as Mendeley, ResearchGate, Academia.edu.

Please see our rights policy on <https://benjamins.com/#authors/rightspolicy>

For any other use of this material prior written permission should be obtained from the publishers or through the Copyright Clearance Center (for USA: www.copyright.com).

Please contact rights@benjamins.nl or consult our website: www.benjamins.com

Exploring absorbing reading experiences

Developing and validating a self-report scale to measure story world absorption

Moniek M. Kuijpers^a, Frank Hakemulder^a, Ed S. Tan^b,
and Miruna M. Doicaru^b

^a Department of Media and Culture, Utrecht University /

^b Amsterdam School of Communication Research, University of Amsterdam

Even though there is extensive research on absorbing experiences with narrative media, an instrument able to measure different aspects of absorption in the story world of a textual narrative has yet to be developed. Such an instrument should be able to predict different evaluative responses while at the same time being sensitive to various stimulus materials, because it could help further the research into the relationships between narrative texts, absorbing experiences and entertainment outcomes. This paper develops such an instrument through a literature review, interview study, pilot study, and exploratory and confirmatory factor analysis. Attention, transportation, emotional engagement and mental imagery were found to be dimensions of story world absorption. The final scale is reliable, sensitive to different stimulus materials and able to predict two different evaluative responses: enjoyment and impact. It is argued that the text a reader reads determines the particular evaluative response to a story world absorption experience.

Keywords: story world absorption, measure, enjoyment, reading, scale development

The experience of feeling absorbed in stories has been studied extensively over the last two decades (see for examples: Nell, 1988; Gerrig, 1993; Green, Brock & Kaufman, 2004; Bilandzic & Busselle, 2008; Busselle & Bilandzic, 2009). Previous research on experiential states of absorption and similar phenomena has developed measures to capture these states and investigate them empirically. Among the available instruments are those that tap into one aspect of an absorbing reading experience, such as Cohen's identification measure (2001) or the suspense and mystery questionnaire of Knobloch and her colleagues (2004) for example. These instruments focus on a specific part of a narrative experience or on certain forms

of narrative. There are also measures that try to capture a multilayered concept of engagement with a narrative (may it be textual or audiovisual), such as Green & Brock's transportation scale (2000) or Busselle & Bilandzic's narrative engagement scale (2009). Even though the validity and usability of these scales has been demonstrated by previous research, these scales do not measure the *subjective experience* of being absorbed in the story world of a *narrative text* and are therefore unsuitable for the purposes of the current study. For example, Busselle & Bilandzic's narrative engagement scale is developed to capture processes that underlie engagement with *audiovisual* media (2009); Green & Brock's transportation scale (2000) does not include items that capture the deictic shift from real world to story world (while the title of the scale seems to imply that it does) and although enjoyment has been investigated as an outcome of absorption before (Tal-Or & Cohen, 2010; Busselle & Bilandzic, 2009), the possibility that it could be a part of absorption has not been ruled out by empirical investigation. Granted, the available scales could be — and have been — adapted for use with different media (Tal-Or & Cohen, 2010; Hall & Zwarun, 2012), combined to capture a multilayered concept of absorption (Hall & Bracken, 2011), or modified to suit different approaches (Balakrishnan & Sundar, 2011). However, it would be preferable to develop an instrument that is specifically designed to measure the subjective experience of being absorbed in the story world of a narrative text during reading. The reason for this is precisely because of the relationship between absorption experiences and enjoyment. The notion that absorption and enjoyment are closely related to each other is an idea prevalent in media psychology (Oliver & Bartsch, 2010; Tal-Or & Cohen, 2010; Green, Brock & Kaufman, 2004; Bilandzic & Busselle, 2008) and has been studied thoroughly. In this study, however, we suggest that we should delve deeper into the question of what the *nature* of this relationship is exactly. Not only should be *empirically* established that enjoyment is indeed an *outcome* of (or response to) an absorption experience rather than an *aspect* of such an experience, we also argue that the text a reader reads determines the particular evaluative response to a story world absorption experience. The instrument that is needed to measure story world absorption should therefore be able to predict different evaluative responses while at the same time being sensitive to various stimulus materials. In that way it could help further the research into the relationships between narrative texts, absorbing experiences and their entertainment outcomes.

This paper develops such an instrument through a literature review that will serve as the foundation for the theoretical dimensions that together form story world absorption. An interview study and pilot study are conducted to phrase and test the items that are part of each of these dimensions. The theoretical construct of story world absorption that results from these studies will be tested and validated using exploratory and confirmatory factor analysis. Apart from that the scale will

be validated in a third study that compares it to two similar scales with correlation analyses, namely the narrative engagement scale of Busselle & Bilandzic (2009) and the transportation scale of Green & Brock (2000). Its ability to predict enjoyment will be tested with regression analyses in all three studies and compared with the predictive ability of the narrative engagement scale and the transportation scale in the last study.

Literature review

The purpose of this paper is to develop an instrument that can capture the multi-layered subjective experience of a readers' absorption in the story world of a textual narrative. This literature review is divided in four paragraphs each devoted to a theoretical dimension of the story world absorption construct. This categorization is based on the observation that a pattern of recurring experiential states can be discerned when reviewing the relevant literature, most importantly Green & Brock's (2000) and Busselle & Bilandzic's work (2009) on scale development. The theoretical dimensions are attention, transportation, emotional engagement and mental imagery. All of these separate aspects can together constitute one story world absorption experience. However, not every reader experiences exactly the same when being absorbed in a story world and no two texts establish exactly the same absorbing experience in different readers. The term story world absorption was chosen because it denotes a focus on the story world of a narrative (and not the other parts of a narrative such as the form) and because the term absorption seems to capture a quality that terms like engagement or involvement cannot reach. Even though the term absorption is known mostly for its use by Tellegen & Atkinson (1974) to denote a character trait, it still feels appropriate to use it, because it can also signify a state and implies a certain power ascribed to the text: 'the story absorbed me'.

After discussing the relevant literatures to the four dimensions mentioned above, one paragraph will discuss enjoyment's relation to story world absorption and a last paragraph will introduce the hypotheses under investigation in this paper.

Attention

Within the construct of story world absorption, attention is characterized by a deep concentration of the reader that *feels* effortless to them. As a consequence the reader can lose awareness of themselves, their surroundings and the elapse of time. While attention can be understood as a character trait (i.e. ability to concentrate, Tellegen & Atkinson, 1974) in the story world absorption construct it refers to the subjective experience a reader has of their own deepened concentration

while being absorbed. Most of the self-report measures that are available consider attention to be part of a larger construct (Busselle & Bilandzic, 2009; Green & Brock, 2000; Csikszentmihalyi, 1988), just like we consider attention to be part of the story world absorption construct. Nevertheless, it differs from these constructs in the sense that it does not refer to a cognitive process underlying understanding (Busselle & Bilandzic, 2009), which could also be assessed by physiological and implicit measures (Ravaja, 2004), rather it refers to the *feeling* readers have of deepened concentration or focus while being absorbed.

Although this may seem like an artificial distinction, we believe it is an important distinction to make when attempting to capture the *felt* experience of readers' absorption and not the underlying processes of that experience. An important assumption about the attention sub-dimension in the story world absorption construct is that readers, while absorbed, are not aware of the cognitive processes that might have instigated their deepened concentration and therefore references to these processes could not be a part of their subjective experience.

Mental imagery

The concept of mental imagery indicates *visual* imagery that comes to mind specifically when reading a story, which can aid a deeper immersion into the story world. The mental imagery dimension of the story world absorption construct is almost identical to the imagery construct of Green & Brock's transportation scale (2000), measuring story world related imagery such as surroundings, characters, and situations that the reader visualizes while reading. Including a mental imagery dimension was necessary, because the story world absorption construct is focused on story worlds in *textual* narratives. Using a mental models approach to investigate engagement with *audiovisual* narratives, Busselle & Bilandzic (2009) did not include a mental imagery dimension, for example. The difference between the two narrative media mainly lies in the way the two media are perceived. When watching a film, viewers are usually presented with more fully formed images of what the characters or the setting look like than readers are when reading a book. The reader is required to use mental capacities to *imagine* what is described in the text, whereas other mental faculties are appealed to in the case of a viewing experience.

Busselle & Bilandzic did, however, incorporate a dimension similar to imagery in their scale (2009), namely *perspective taking*, to accommodate the audiovisual stimuli they work with. Perspective taking or 'sharing the perspective of the character' is also part of Cohen's construct of identification with fictional characters (2001). It could be argued that perspective taking is part of the mental imagery process, if readers place themselves in the position of the protagonist in the fictional world.¹ Film viewers are, by the nature of the medium, forced into a certain

perspective or viewpoint in the film, whereas readers have more freedom in that respect.

Emotional engagement

Emotional engagement as an aspect of the story world absorption construct includes feelings *for* or *with* characters, such as sympathy, and empathy, and identification. This dimension does not account for overall emotional responses to the whole story (such as: this story was emotional or this story made me emotional); it only encompasses story related, or rather *character* related emotional engagement. As with research into attention, the research into emotional engagement is extensive and covers studies on emotional engagement as a trait (the Literary Response Questionnaire of Miall & Kuiken, 1995), as part of a larger construct (Green & Brock, 2000; Busselle & Bilandzic, 2009) or even as a construct that *includes* absorption (the identification scale of Cohen, 2001).

Certainly, we can ask ourselves, as Cohen does, how the two are related: do people first get emotionally involved and thereby absorbed or do they get absorbed first which clears the way to get emotionally involved? The research of both Green & Brock (2000) and Busselle & Bilandzic (2009) seems to solve this problem by incorporating emotional engagement in a larger construct of absorption and not looking at the two concepts as separate and causal. Tal-Or & Cohen (2010) also focus on the relationship between identification and transportation. The authors ask themselves a similar question: are the two causally related, are they synonymous, or separate, but similar concepts? Through the use of exploratory factor analyses, Tal-Or & Cohen (2010) were able to distinguish between the two concepts. In the case of the story world absorption construct, a distinction is made between emotional engagement and transportation as two parts of story world absorption and therefore they are separate concepts but can work together to generate absorption. However, it seems possible that, in reading different texts, sometimes the two co-occur while at other times one is much stronger than the other, perhaps even causing the other.

Transportation

In the story world absorption construct, transportation signifies a feeling of entering a story world, without completely losing contact with the actual world. The term transportation was chosen to describe this experience, since it describes best the movement from actual world to story world, while terms such as presence do not refer to that movement. In the relevant literature many similar concepts are discussed. All of these concepts are related to each other and to Gerrig's original

concept of transportation or *experiencing narrative worlds* (1993). In phenomenological terms, transportation denotes the feeling of being in the story world and consequently away from the actual world (Gerrig, 1993). Following Gerrig, Green & Brock consider transportation as ‘a distinct mental process, an integrative melding of attention, imagery and feelings’ (2000, 701). Green & Brock combine the three aspects mentioned above into one construct they call transportation. However, in other conceptualizations — such as narrative presence (Busselle & Bilandzic, 2009), aesthetic illusion (Wolf, 2004) or deictic shift (Duchan, Bruder & Hewitt, 1995; Segal, 1995) — transportation is an experience that exists separate of these three aspects. Where Wolf describes aesthetic illusion as the ‘feeling of being recentered in a possible world’ (2004, 325), Busselle & Bilandzic claim that narrative presence — ‘the sensation of being present in a narrative world’ — is the result of comprehension processes and perspective taking (2009). Both Wolf’s and Busselle & Bilandzic’s ideas are closely related to deictic shift theory, which focuses on the ‘psychological relocation [of the reader] into the story’ (Duchan, Bruder & Hewitt, 1995; Segal, 1995). These concepts all differ from tele-presence — a related concept introduced by Lee (2004) in the field of game studies — in the sense that tele-presence is much more stringent than transportation (and related terms): it assumes that the reader (or recipient) is completely *present* in the fictional world and no longer in the actual world. In the case of the other constructs — deictic shift, narrative presence and aesthetic illusion — contact with the actual world remains intact (Segal, 1995; Busselle & Bilandzic, 2009; Wolf, 2004). This is also true for transportation as part of the story world absorption construct. Furthermore, transportation in the present study is seen as existing alongside emotional engagement, mental imagery and attention, but obviously closely related to all three. Apart from that, it is not understood as a cognitive process, as it is in the research of Busselle & Bilandzic (2009) for example, but rather as the subjective experience readers feel of having been to the story world while they were absorbed.

Enjoyment

It has been hypothesized and investigated before that narratives that are absorbing or engaging result in greater enjoyment (Green, Brock & Kaufman, 2004) and after effects such as story-related attitudes (Bilandzic & Busselle, 2008; Busselle & Bilandzic, 2009) or changes of world beliefs (Green & Brock, 2000). In fact, it has been argued that the whole reason people engage in reading stories is to enjoy themselves (Brewer & Lichtenstein, 1982). Enjoyment has been conceptualized as an attitude (Nabi & Krcmar, 2004), as an absorbing experience in itself (Green, Brock & Kaufman, 2004), as an outcome of an absorbing experience (Oliver & Bartsch, 2010) and as the satisfaction of intrinsic needs (Tamborini, Bowman,

Eden, Grizzard, & Organ, 2010). There are also definitions of enjoyment that seem to encompass elements of all these conceptualizations and that are actually very similar to definitions of absorption. For example, both enjoyment and absorption have been characterized as ‘an intrinsically rewarding experience or activity that is sought by [readers] independently of extrinsic rewards’ (Oliver & Bartsch, 2010).

To summarize, it seems the relevant literature is not clear on the exact nature of the relationship between absorption and enjoyment. It could be argued that enjoyment can be felt *during* an absorption experience, as well as *after* the experience. The latter is considered the most probable by the relevant literature (Tal-Or & Cohen, 2010; Green, Brock & Kaufman, 2004) and is (empirically) researched quite extensively. However, the former has not been empirically investigated before. In this paper, two exploratory factor analyses will be performed — one including enjoyment items and one excluding them — to examine whether enjoyment should indeed be considered an outcome rather than a part of story world absorption or not.

Hypotheses

In sum, story world absorption can be defined as an experiential state that can emerge during the reading of a narrative text. This state is characterized by readers’ focused attention on the alternate story world presented in the text — as a consequence of which readers can become less aware of their surroundings and themselves and can lose track of time; during reading readers can feel transported to the world of the story and this feeling can be supported by strong emotional reactions to what happens in the story world and by mental imagery readers generate of the story world. There is a possibility that there are more aspects to the subjective experience of story world absorption that are overlooked so far, however these four categories seem to cover the spectrum of relevant literature on the topic and coincide with what was found in the interview study presented further below. This research covers the same bases, in terms of sub-dimensions, as Green & Brock (2000) and Busselle & Bilandzic (2009) and therefore can be seen as extending the research that they conducted on this topic. The hypotheses that are the focus point of the present studies are the following:

- H1: Story world absorption consists of four dimensions, namely attention, emotional engagement, mental imagery, and transportation
- H2: The type of text that is read determines the story world absorption score: texts that make use of more popular techniques, such as suspense, will lead to higher story world absorption scores, while texts that make use of more literary techniques, such as foregrounding, will lead to lower absorption scores

H3: The content of a text that is read determines the story world absorption score: action-oriented stories will lead to higher story world absorption scores than character-oriented stories

H4: Enjoyment is an evaluative response that can be predicted by story world absorption, but is not part of it.

Overview of studies

The development of the story world absorption scale (SWAS) progressed through five different studies with five different data sets and different sets of stimulus material. The process began with a small interview study (N=8) to inspire the formulation of the items used in the exploratory factor analysis (EFA). A pilot study followed (N=59) to test the understandability of the items. The final list with items was then submitted to EFA with a third data set (N=278). Confirmatory factor analysis (CFA) was performed with a fourth data set (N=129). Both in the EFA and in the CFA study the scale's ability to predict enjoyment was assessed with regression analyses. Finally, in a fifth study the scale's relations to related scales, namely the narrative engagement scale (Busselle & Bilandzic, 2009) and the transportation scale (Green & Brock, 2000) was investigated to assess the convergent validity of the SWAS and compare the three scales' abilities to predict enjoyment.

In the following sections the interview study and the pilot study are briefly discussed first. Then under the heading of study 1 the EFA will be discussed and under the heading of study 2 the CFA will be discussed. After that, under the heading of study 3, the convergent validity analysis and comparative predictive validity analyses are briefly discussed. Each of these sections first presents the results from the scale development or validation studies and then the results from the regression analyses.

Interview study

For the purpose of articulating the items on the SWAS eight respondents (two male and six female) from a convenience sample were interviewed about their reading experiences. Their ages ranged from 23 to 53 with a mean age of 31.20 and a standard deviation of 11.61. The purpose of these interviews was mainly to gather data from actual readers about how they *express* feelings of absorption in narrative texts. The choice for either devising new items or adapting items from existing scales depended on the interview study. If the interviewees spoke of their reading experiences in terms completely different from those used in other scales, new items were created based on the interviewees' expressions. Otherwise items were adapted from other scales, such as the identification measure of Cohen (2001), the

transportation scale of Green & Brock (2000) and the narrative engagement scale of Busselle & Bilandzic (2009).

Before we started with the interview, the participants were shortly briefed about the purpose of the interview. They were told that this research is about reading experiences, that they could feel free to talk about their experiences, and that their anonymity was ensured. Since these interviews were conducted for exploratory reasons an open interview structure was used. A list of basic open-ended questions guided the interviews, but the interviewees were encouraged to associate and let their minds wander over their personal reading experiences.

The interview study seemed to confirm the tentative structure of the story world absorption construct of four dimensions: attention, emotional engagement, mental imagery and transportation. Additionally, items were included to capture value judgments of readers, since they were mentioned often by the interviewees and it was not completely clear whether the evaluative responses should be included in the absorption experience itself or whether they could be seen as outcomes of absorption experiences. Table 1 presents a full list of the items categorized per dimension with references to earlier scales.

Pilot study

For each theoretical dimension a couple of items (an average of seven items) was formulated based on the expressions of the interviewees. Whenever their expressions overlapped with an item on a previous scale, the original item from that scale was adapted, translated, and added to the dimension it seemed to belong to. Prior to the exploratory factor analysis, a group of 59 participants (literature and art students of the Universities of Utrecht and Groningen) in a pilot study were asked to read either one of two stories (*De Toren* [translation: *The Tower*] by Bas Heijne and *De Invaller* [translation: *The Substitute*] by René Appel), fill in the questionnaire and relate whether or not they found the questionnaire understandable. The ages of the participants ranged from 18 to 27 with a mean of 21.25 and a standard deviation of 2.30. Eleven of the participants were male and 48 of the participants were female. After the study they were asked to give their opinions on the questionnaire; whether it was understandable and clear what they had to do. Negative formulations of the items proved difficult to understand for the participants and were thus excluded from the scale prior to the EFA. New items replaced three items on the transportation dimension, because they seemed to measure presence more than transportation and because the participants claimed not to experience presence at all.

Table 1. Items used for developing the story world absorption scale

Dimension: Attention
A1: The story gripped me in such a way that I could close myself off for things that were happening around me.
A2: When I was reading the story I was focused on what happened in the story.
A3: When I finished the story I was surprised to see that time had gone by so fast. (Adapted from Busselle & Bilandzic, 2009)
A4: I was reading in such a concentrated manner that I had forgotten the world around me.
A5: I was not that aware of myself while I was reading the story.
A6: Because all of my attention went into the story, I sometimes felt as if I could not exist separate from the story.
Dimension: Emotional engagement
EE1: I felt connected to the main character in the story.
EE2: I felt how the main character was feeling.
EE3: At certain key moments in the story, I felt exactly what the characters were feeling. (Adapted from Cohen, 2001)
EE4: I felt sympathy for the main character.
EE5: I was worried about the main character in the story. (Adapted from Cohen, 2001)
EE6: I felt sorry for the main character in the story. (Adapted from Cohen, 2001)
EE7: The story touched me in a certain way. (Adapted from Cohen, 2001)
EE8: I felt for what happened in the story.
EE9: I thought it was an emotional story.
Dimension: Mental Imagery
MS1: When I was reading the story I had an image of the main characters in mind. (Adapted from Green & Brock, 2000)
MS2: When I was reading the story I could see the situations happening in the story being played out before my eyes. (Adapted from Green & Brock, 2000)
MS3: I could imagine what the world in which the story took place looked like.
MS4: When I read the story I could imagine what it must be like to be in the shoes of the main character. (Adapted from Cohen, 2001)
MS5: The story made me think of something that happened to me.
MS6: I saw what happened in the story through the eyes of the main character.
Dimension: Transportation
T1: Something in the story stuck with me after I finished reading it.
T2: When reading the story there were moments in which I felt that the story world overlapped with my own world.
T3: When I was finished with reading the story it felt like I had taken a trip to the world of the story.
T4: When I was reading the story it sometimes seemed as if I were in the story world too.
T5: The world of the story sometimes felt closer to me than the world around me. (Adapted from Busselle & Bilandzic, 2009)
T6: When I look back it seemed as if the beginning of the story had lifted me out of the real world and at its end I returned into the real world.
Dimension: Enjoyment
E1: I thought it was an exciting story.
E2: I rather did not want the story to end.
E3: I was constantly curious about how the story would end. (adapted from Green & Brock, 2000)
E4: I thought it was a compelling story.
E5: I thought it was fun to read this story.
E6: I thought it was an enthralling story.
E7: I read the story with great interest.
E8: I thought the story was beautiful.
E9: I want to read the story again sometimes.
E10: I thought the story was written well.
E11: The style in which the story was written was appealing to me.
E12: I thought it was very rewarding to read this story, because the story was complex at times.
E13: I thought this story was special.
E14: I thought the style of this story was striking.
Shared notion of absorption with researchers
AB1: I felt completely occupied by the story.
AB2: I felt absorbed in the story. (Adapted from Busselle & Bilandzic, 2009)

Study 1: Scale development

Method

Aims. This study was designed to develop a new scale that measures story world absorption. Apart from that, this study purported to find out whether this scale is sensitive to different materials to ensure its validity by conducting a multivariate ANOVA on both stories. Another aim of this study was to find out whether and

how story world absorption is related to enjoyment and to ensure the predictive validity of the newly developed scale.

Participants. 86 first year students from the University of Groningen (literature and film students) and from the University of Utrecht (social sciences) and 192 senior high school students participated in this study voluntarily. Their ages ranged from 15 to 36, with a mean of 18.19 and a standard deviation of 2.58. 148 participants were female and 127 participants were male (3 participants failed to fill out their gender).

Design. This study employed a two-factor design. One group read *The Tower* and the other group read *The Substitute* and both received the same instructions and questionnaire. No text manipulations were applied, since the main aim was to test and validate the scale.

Materials. Two different Dutch short stories (*De Toren* [translation: *The Tower*] by Bas Heijne and *De Invaller* [translation: *The Substitute*] by René Appel) were randomly assigned to the participants. The stories are both approximately two pages long, *The Tower* consists of 1006 words and *The Substitute* consists of 744 words. *The Tower* is a story that focuses more on the inner life of one of its main characters, while *The Substitute* is a story with a strong focus on actions and events that happen in the story. In concurrence with relevant literatures (Tan & Fasting, 1996; Vorderer 1996) it is assumed that action-oriented stories will elicit higher story world absorption scores than character-oriented stories. Therefore it is expected that *The Substitute* will cause higher story world absorption scores.

Measures. To measure story world absorption 43 items were developed for the first version of the SWAS. These items were randomized on the questionnaire to prevent item order from influencing factor structure. Participants had to rate their experience with the story on a 7-point Likert rating scale ranging from -3 (completely disagree) to 3 (completely agree). Because the pilot study demonstrated that reverse-scored items proved difficult to understand for the participants there are no such items included in the scale. This choice was made to avoid the risk of moving participants further away from their original experience. Table 1 displays all of the items that were included in the EFA. Five extra items measuring understandability were added, to control for understandability of the stories used.

Procedure. The participants received instructions on how to read the story (“read the story as you would read a book at home for your own pleasure”) and fill out the questionnaire. They were seated in a classroom and were asked to leave the room quietly so as not to disturb the other participants when they were done. They were then asked to read one of the two stories, which were randomly distributed among them. After that they had to fill out the questionnaire and mark their age and gender on the form as well (to control for within-group variance).

Results

First an EFA was conducted on all of the items (including the enjoyment items) and after that another EFA was conducted including only the theoretical dimensions of story world absorption (without enjoyment items). Then the results of the two structures were compared and the best structure (explaining more variance and with higher factor loadings) was chosen to work with in the CFA in study 2. The two EFA's and their comparison will be discussed first and after that the differences in story world absorption scores between the two stories, assessed through the use of ANOVA's, will be discussed. Finally, two regression analyses will be discussed that test the predictive validity of the newly developed measure.

First exploratory factor analysis. Of the 278 questionnaires that were filled out, 8 were discarded before the analysis, because of missing values. Exploratory factor analysis was used to identify items that define one dimension; to see if these dimensions correspond with the dimensions in the theory presented above; and to obtain a smaller number of items for further use in the questionnaire. Four rounds of principal components analyses were run, each time using oblique rotation. Oblique rotation was used as opposed to orthogonal rotation, because the theory suggests that the factors derived from the exploratory factor analysis might be correlated (Field, 2005, p. 645). The selection criterion for retaining items was a primary loading greater than .45. Every item that had a secondary loading that was greater than half of its primary loading had to be eliminated as well, since secondary loadings may indicate problematic items. The Kaiser-Meyer-Olkin measure in the first round was .944, indicating that the data set was very well suited for factorability. Overall, the eight-factor structure that was derived in the first round might have been overestimated, as suggested by the size of the factors and the scree plot. Two variables were excluded, because communalities showed that they did not correlate highly enough with any of the other variables. After the second round all of the variables that did not load on to any factors and the variables that did not correlate highly enough were excluded. One factor was eliminated and a seven-factor structure was retained, which was, however, still overestimated. Therefore two more analyses requesting first a five-factor solution and then a six-factor solution were carried out. Because there were several factors with one or two variables loading lower than .45 in the five-factor solution and the six-factor solution had only one factor that contained a variable loading lower than .45, the six-factor solution was more satisfactory.

Before rotation the six factors in the six-factor solution explained 63 percent of common variance and met Kaiser's criterion of eigenvalues greater than 1.0. Of the 38 items left there was only one item that had a factor loading of below .45 and there were 4 items that double loaded. The sum of squared loadings after rotation

suggested that all factors were of roughly the same importance. Interpretability was considered to be more important than percentage of variance and therefore as many factors as possible were retained. The factor analysis was repeated omitting the variables that did not load or that double loaded in the six-factor solution. All factors remained stable in two runs. During the first run one more item was omitted, because it did not load onto any factor.

The analysis yielded six dimensions, four of which seemed to reflect aspects of the story world absorption experience. The first, labeled *attention* contains focused attention and general absorption items and describes readers' concentration during reading. The third factor, *emotional engagement*, describes reader's emotional engagement with the characters in the story. The fourth factor labeled *mental imagery* describes story related imagery that readers have in mind while reading the story. The sixth factor is labeled *transportation* and describes the feeling of being transported to the story world (and back to the real world). The second and fifth factor differed from the other four in the sense that they contained items with value judgments from the reader. One factor, *enjoyment*, pertains more to the readers' general enjoyment of what they have read and the other factor, *impact*, includes items on the emotional impact the story had on the readers. This last factor was the least clear and its reliability as a separate scale was rather low ($\alpha = .61$). Table 2 contains a table with factor loadings, communalities, the sum of squared loadings after rotation, percentage of variance before rotation and correlations among the factors and Cronbach's α for each of the separate factors (Cronbach's α for the whole scale was .84 which indicates that all of the items together form a consistent scale). These statistics apply to the 30 variable, six-factor solution that includes enjoyment items.

Second exploratory factor analysis. Since exploratory factor analysis entails simple structure dimensional separation, the relationships between story world absorption (the four-factor structure consisting of attention, emotional engagement, mental imagery and transportation) and the evaluative responses (the two-factor structure of enjoyment and impact) might be attenuated in the six-factor structure. Therefore, the four-factor structure of story world absorption without enjoyment was assessed with EFA as well. This also allowed for an examination of enjoyment's (and impact's) relationship to story world absorption: the comparison can tell us whether enjoyment and impact are indeed outcomes of absorption rather than aspects of the experience. A principal components analysis with direct oblimin rotation was performed with the twenty items that comprise the story world absorption construct. The Kaiser-Meyer-Olkin measure was .922 indicating that the data was very well suited for factorability. The four factors together explained 65 percent of variance and met Kaiser's criterion of eigenvalues greater than 1.0. Table 3 contains a table with factor loadings, communalities, the

Table 2. Factor loadings and communalities for the six factor structure EFA

Original items	Factor						<i>h</i> ²
	1	2	3	4	5	6	
A3	.75						.64
A2	.70						.63
A4	.69						.80
AB2	.62						.69
A1	.60						.67
A5	.58						.45
E11		-.88					.73
E5		-.82					.82
E6		-.77					.79
E10		-.76					.64
E7		-.63					.73
E8		-.61					.65
E1		-.48					.57
MS4			-.76				.68
EE4			-.73				.77
EE1			-.63				.48
EE2			-.62				.68
EE5			-.60				.63
EE8			-.57				.71
MS3				.80			.67
MS2				.76			.69
MS1				.74			.58
EE9					.74		.56
E13					.73		.65
E12					.64		.53
T4						-.73	.68
T2						-.69	.55
T5						-.66	.72
T3						-.57	.71
A6						-.56	.61
Sum of squared loadings after rotation	6.64	7.08	6.44	3.68	3.75	5.44	
Percentage of variance before rotation	38.23	8.00	5.56	5.25	4.51	4.03	
Cronbach's α	.87	.91	.88	.68	.61	.85	
Correlation factor 1	1.00						
Correlation factor 2	-.35	1.00					
Correlation factor 3	-.34	.36	1.00				
Correlation factor 4	.28	-.23	-.22	1.00			
Correlation factor 5	.22	-.32	-.23	.13	1.00		
Correlation factor 6	-.36	.26	.33	-.21	-.22	1.00	

sum of squared loadings after rotation, percentage of variance before rotation and correlations among the factors and Cronbach's α for each of the separate factors

Table 3. Factor loadings and communalities for the four factor structure EFA

Original items	Factor				h^2
	1	2	3	4	
A3	.78				.59
A4	.73				.79
A2	.71				.62
AB2	.68				.70
A5	.68				.44
A1	.67				.64
EE4		-.86			.78
MS4		-.74			.65
EE1		-.72			.51
EE8		-.70			.72
EE5		-.69			.59
EE2		-.67			.66
MS3			.83		.69
MS2			.78		.69
MS1			.74		.56
T4				.70	.68
T2				.66	.56
T5				.63	.72
T3				.56	.70
A6				.53	.62
Sum of squared loadings after rotation	6.12	5.99	3.45	4.26	
Percentage of variance before rotation	42.42	8.09	7.78	6.38	
Cronbach's α	.87	.88	.68	.85	
Correlation factor 1	1.00				
Correlation factor 2	-.46	1.00			
Correlation factor 3	-.33	-.31	1.00		
Correlation factor 4	.35	-.36	.16	1.00	

(Cronbach's α for the whole scale was .92. These statistics apply to the final 20 variable, four-factor solution.

Comparison. To determine which structure proves to be a better model for use with confirmatory factor analysis, a comparison of the results is required. Overall, the six-factor structure explained 63 percent of variance, whereas the four-factor structure explained 65 percent of variance. The factor loadings were generally higher in the four-factor structure (with the exception perhaps of the

transportation sub-dimension), as were the correlations between factors. When comparing Table 2 and 3 a few other things stand out. The evaluative items all loaded onto either enjoyment or impact; there were no 'cross overs' to any of the other four dimensions. In other words: there is a clear split between items that belong to story world absorption and items that belong to evaluative responses. Also, the dimensions of the story world absorption construct were exactly the same in both analyses, which gives confidence in stating that it is a stable construct. The choice was then made to work with the four-factor structure of story world absorption in the CFA, because it seems that excluding the evaluative items from the EFA resulted into a better structure accounting for more variance with fewer items.

Analyses on sensitivity and predictive validity of the SWAS. To assess the sensitivity of the SWAS to different materials and answer the hypotheses concerning text differences, a multivariate ANOVA was conducted with story world absorption and its four separate dimensions as dependent variables, condition as independent variable and age and gender as covariates. A significant difference was found between the two stories and their overall story world absorption scores, $F(1,266) = 7.74$, $p = .006$, $\eta_p^2 = .168$. The group of participants who read *The Substitute* ($M = 4.54$ $SD = 1.04$ $n = 136$) scored significantly higher on story world absorption than the group who read *The Tower* ($M = 4.21$ $SD = 1.00$ $n = 134$). This was in line with the expectation that action-oriented stories would elicit more story world absorption. When looking closer at the four separate dimensions, the only difference that was not significant was the difference in mental imagery scores (*The Substitute*: $M = 5.94$ $SD = .98$ / *The Tower*: $M = 5.84$ $SD = .99$). In the other three cases, the group who read *The Substitute* scored significantly higher: on transportation (*The Substitute*: $M = 3.53$ $SD = 1.40$ / *The Tower*: $M = 3.17$ $SD = 1.27$), on emotional engagement (*The Substitute*: $M = 4.48$ $SD = 1.29$ / *The Tower*: $M = 4.08$ $SD = 1.29$) and attention (*The Substitute*: $M = 4.77$ $SD = 1.28$ / *The Tower*: $M = 4.38$ $SD = 1.26$).

In order to assess the predictive validity of the SWAS, its relation with enjoyment and impact was explored. Enjoyment correlated significantly (at the .01 level) with the overall SWAS (*The Substitute*: .609 / *The Tower*: .671) as well as impact (*The Substitute*: .451 / *The Tower*: .481). Two separate regression analyses, one for each of the two stories, were then conducted. Enjoyment and impact were both regressed onto the four sub-scales of the SWAS. The results of these analyses can be found in Table 4 for *The Tower* and Table 5 for *The Substitute*. The data suggest that together the four sub-scales are a good predictor of enjoyment (*The Substitute*: 43 percent / *The Tower*: 46 percent) and to a lesser extent of impact as well (*The Substitute*: 23 percent / *The Tower*: 27 percent). The data also suggest, however, that the sub-scales do not contribute to enjoyment or impact equally. In both conditions, emotional engagement and attention are the strongest predictors of enjoyment, however only emotional engagement contributes significantly

to both enjoyment and impact in the case of *The Tower* and only attention contributes significantly to both enjoyment and impact in the case of *The Substitute*.

Table 4. Results of regression analysis on enjoyment and impact for *The Tower* (N = 134)

	Enjoyment	Impact
Transportation	.16	-.03
Emotional Engagement	.39***	.39***
Mental Imagery	.01	.06
Attention	.23**	.18
R²	.46	.27
F	27.31***	12.12***

* = $p < .05$, ** = $p < .01$, *** = $p < .00$

Table 5. Results of regression analysis on enjoyment and impact for *The Substitute* (N = 136)

	Enjoyment	Impact
Transportation	-.13	.21
Emotional Engagement	.33***	.08
Mental Imagery	.09	-.03
Attention	.45***	.27*
R²	.43	.23
F	24.88***	9.72***

* = $p < .05$, ** = $p < .01$, *** = $p < .00$

Study 2: Scale validation

Method

Aims. This study was designed to test the 20-item SWAS that was derived from the exploratory factor analyses in study 1, with confirmatory factor analysis. The hypothesis that was tested with this analysis was that story world absorption consists of four different dimensions: transportation, attention, emotional engagement and mental imagery. Apart from that, regression analyses again tested the predictive ability of the SWAS.

Participants. A group of 129 students from the University of Utrecht filled out the questionnaire. The questionnaire was brought to their attention via different

Story World Absorption	(.95) Attention	(.78) A3
		(.70) A2
		(.85) A4
		(.92) AB2
		(.81) A1
	(.91) Emotional Engagement	(.86) MS4
		(.84) EE4
		(.74) EE1
		(.90) EE2
		(.80) EE8
	(.64) Mental Imagery	(.86) MS3
		(.80) MS2
		(.69) MS1
	(.95) Transportation	(.84) T4
		(.67) T2
		(.80) T5
		(.84) T3
		(.81) A6

Figure 1. CFA model of story world absorption with higher order latent variable

social media such as Facebook or LinkedIn. Their ages ranged from 20 to 35, with a mean of 26.08 and a standard deviation of 4.21. 97 participants were female and 32 participants were male. All of them were selected on their HBO or WO level education (Dutch higher education).

Design. This study employed a two-factor design, in which the two conditions were determined by the stories the participants read. One group read one story and the other group read another story and both received the same instructions and questionnaire. No text manipulations were applied, since the main aim was to test and validate the scale.

Materials. Two different short stories were randomly assigned to the subjects. A Dutch translation of Hemingway’s short story *Hills Like White Elephants* translated by Clara Eggink (*Heuvels als Witte Olifanten*) and a Dutch short story by L.H. Wiener: *Het Satanisch Geheugen* [translation: *Satanic Memory*]. The stories are both about two to three pages long. *Hills Like White Elephants* consists of 1452 words and *Satanic Memory* consists of 1068 words. While *Hills Like White Elephants* is more descriptive in terms of the setting, *Satanic Memory* is more descriptive in terms of the inner life of the protagonist. In line with the hypotheses

in the first study, it is expected that *Satanic Memory* will elicit significantly higher emotional engagement scores than *Hills Like White Elephants*, because it is focused more on character. *Hills like White Elephants* is not really focused on action or character, but does give many descriptions, therefore it is expected that it will elicit significantly higher mental imagery scores than *Satanic Memory*. However, overall it is expected that *Satanic Memory* will elicit higher story world absorption. Both stories are more literary than the two stories used in the first study. *Hills Like White Elephants* makes use of an unusual perspective without insight in the characters feelings and motives, whereas *Satanic Memory* is a story with an unusual topic that uses a lot of metaphor. Determined by foregrounding analyses on the four stories, carried out in the same fashion as Miall & Kuiken (1995), both stories use a higher level of deviation than the two stories used in the first study. The term foregrounding denotes a defamiliarizing experience with literary texts, usually achieved by diverting the attention of the reader to the form of representation of a story and away from the story itself (Hakemulder, 2004). Since a shift *away* from the story world seems detrimental to story world absorption it is expected that the two stories used in this study elicit lower story world absorption scores than the stories used in the first study.

Measures. For the confirmatory factor analysis the questionnaire was adapted into an online 20-item version. Again these items were randomized to prevent item order from influencing factor structure. The items that were used to measure enjoyment and impact in the first study were used again in this experiment. Five extra items measuring understandability were added, to control for understandability of the stories that were used.

Procedure. The participants were asked to read one of the two stories, which were randomly distributed among them. After that they had to fill out the adapted 20-item SWAS and mark their age and gender on the form as well (to control for within-group variance). Also, they were asked to fill out questions about enjoyment and impact of the story they read. Since the participants had to fill out the questionnaire online a few extra control questions were added, such as questions about their level of education, which studies they were enrolled in and whether or not they had participated in this study before.

Results

Confirmatory factor analysis. The model that was tested using CFA was based on the outcome of the exploratory factor analyses and therefore consisted of four factors that vary in size. The model includes one higher order latent variable representing story world absorption. The reason to include this higher order latent variable was that such a model would not only allow the evaluation of the factor

structure, but also the relation of the factors to story world absorption. The evaluation of the model was based on overall goodness of fit tests: chi-square to degrees of freedom ratio, the CFI and RMSEA tests, inspection of individual path coefficients, and modification indices.

First the data set was inspected for missing values and non-normality. Twenty-three participants were removed from the sample because of missing values, leaving 106 participants in the sample. The assessment of multivariate normality tests showed non-normality for a couple of the items and therefore the Bollen-Stine bootstrap was performed to control for non-normality. The goodness of fit tests indicated that the 20 item model was a moderate fit, but could be improved by modification: $\chi^2=264,034$, $df=166$, $\chi^2/df=1,591$, CFI=.937, RMSEA=.075 (range=.058 to .092). The χ^2 being rather high and significant would usually lead to the interpretation of the model as a bad fit, however since the data were corrected by the bootstrap for non-normality, the significance level of the Bollen-Stine Bootstrap was examined ($p=.075$), which indicated that the model was a moderately good fit. Since the modification indices suggested that there were issues with both the attention and the emotional engagement factors, the item with the lowest beta coefficient from each factor was removed (A5 from attention and EE5 from emotional engagement). This resulted in a four-factor structure with three factors containing five items and one factor containing three items. The fit values increased slightly after removing the two items mentioned above: $\chi^2=211,344$, $df=131$, $\chi^2/df=1,613$, CFI=.945, RMSEA=.076 (range=.057 to .095). The Bollen-Stine Bootstrap was non-significant ($p=.065$) indicating a moderately good fit. The modification indices suggested that the model fit could still be improved by modification, however additional changes that would have improved the model fit would leave a theoretically unsatisfying model with too few items per factor too be useful. All of the beta coefficients in the final model are significant at the .001 level and can be found in Figure 1.

Table 6 shows all of the items that remained in the final four-factor structure of story world absorption and the reliability scores per sub-scale (Cronbach's α for the overall scale: .954).

Analyses on sensitivity and predictive validity of the SWAS. To assess the sensitivity of the SWAS to different materials and answer the hypotheses concerning text differences, a multivariate ANOVA was conducted with story world absorption and its four separate dimensions as dependent variables, condition as independent variable and age and gender as covariates. A significant difference was found between the two stories and their overall story world absorption scores, $F(1,125)=25.29$, $p=.000$, $\eta_p^2=$. The group of participants who read *Satanic Memory* ($M=4.12$ $SD=1.29$ $n=46$) scored significantly higher on story world absorption than the group who read *Hills Like White Elephants* ($M=2.98$ $SD=1.09$

Table 6. A list of the final items retained on the story world absorption scale (the codes for the items do not correspond with the codes for the original items)

Attention (α : .905)

A1: When I finished the story I was surprised to see that time had gone by so fast.

A2: When I was reading the story I was focused on what happened in the story.

A3: I felt absorbed in the story.

A4: The story gripped me in such a way that I could close myself off for things that were happening around me.

A5: I was reading in such a concentrated matter that I had forgotten the world around me.

Transportation (α : .902)

T1: When I was reading the story it sometimes seemed as if I were in the story world too.

T2: When reading the story there were moments in which I felt that the story world overlapped with my own world.

T3: The world of the story sometimes felt closer to me than the world around me.

T4: When I was finished with reading the story it felt like I had taken a trip to the world of the story.

T5: Because all of my attention went into the story, I sometimes felt as if I could not exist separate from the story.

Emotional Engagement (α : .914)

EE1: When I read the story I could imagine what it must be like to be in the shoes of the main character.

EE2: I felt sympathy for the main character.

EE3: I felt connected to the main character in the story.

EE4: I felt how the main character was feeling.

EE5: I felt for what happened in the story.

Mental Imagery (α : .795)

MS1: When I was reading the story I had an image of the main character in mind.

MS2: When I was reading the story I could see the situations happening in the story being played out before my eyes.

MS3: I could imagine what the world in which the story took place looked like.

$N=83$). When looking closer at the four separate dimensions, again the only difference that was not significant was the difference in mental imagery scores (*Satanic Memory*: $M=5.12$ $SD=1.28$ / *Hills Like White Elephants*: $M=4.72$ $SD=1.55$). In the other three cases, the group who read *Satanic Memory* scored significantly higher: on transportation (*Satanic Memory*: $M=3.37$ $SD=1.43$ / *Hills Like White Elephants*: $M=2.25$ $SD=1.18$), on emotional engagement (*Satanic Memory*: $M=4.38$ $SD=1.44$ / *Hills Like White Elephants*: $M=2.65$ $SD=1.22$) and attention (*Satanic Memory*: $M=4.00$ $SD=1.55$ / *Hills Like White Elephants*: $M=2.98$ $SD=1.29$).

The hypothesis that the second two stories would bring about lower SWAS scores than the stories from the first study can only partly be supported (*Hills Like White Elephants*: $M=2.98$ $SD=1.09$ / *Satanic Memory*: $M=4.12$ $SD=1.29$ / *The Tower*: $M=4.21$ $SD=1.00$ / *The Substitute*: $M=4.54$ $SD=1.04$). The stories in the second study did indeed instigate lower scores than the stories in the first studies; however, since two different samples were used it is not possible to compare the scores through an ANOVA.

To assess the predictive validity of the SWAS its relations with enjoyment and impact were explored. Enjoyment correlated significantly (at the .01 level) with

the overall SWAS (*Hills Like White Elephants*: .858 / *Satanic Memory*: .811), as did impact (*Hills Like White Elephants*: .600 / *Satanic Memory*: .730). Two separate regression analyses were then conducted for each of the two stories. Enjoyment and impact were both regressed onto the four sub-scales of the SWAS. The results of these analyses can be found in Table 7 for *Hills Like White Elephants* and Table 8 for *Satanic Memory*. The data suggest that together the four sub-scales are a good predictor of both enjoyment (*Hills Like White Elephants*: 77 percent / *Satanic Memory*: 66 percent) and impact (*Hills Like White Elephants*: 47 percent / *Satanic Memory*: 61 percent). However, the data also suggest that the sub-scales do not contribute to enjoyment or impact equally. The most important difference being that transportation does contribute significantly to enjoyment in both stories, but not to impact.

Table 7. Results of regression analysis on enjoyment and impact for *Hills Like White Elephants* (N=83)

	Enjoyment	Impact
Transportation	.20*	.09
Emotional Engagement	.46***	.66***
Mental Imagery	.02	-.11
Attention	.28**	.01
R ²	.77	.47
F	63.77***	17.25***

* = p <.05, ** = p <.01, *** = p <.00

Table 8. Results of regression analysis on enjoyment and impact for *Satanic Memory* (N=46)

	Enjoyment	Impact
Transportation	.31*	-.15
Emotional Engagement	.21	.46*
Mental Imagery	.04	-.03
Attention	.33	.49*
R ²	.66	.61
F	20.22***	16.12***

* = p <.05, ** = p <.01, *** = p <.00

Study 3: Scale validation

Method

Aims. To assess the convergent validity of the SWAS, a correlation analysis was carried out among the SWAS, the narrative engagement scale (Busselle & Bilandzic, 2009) and the transportation scale (Green & Brock, 2000). Since both the scales of Busselle & Bilandzic and Green & Brock purport to measure a similar experience to the one that the SWAS tries to capture, high correlations between these three scales and their sub-scales are expected. Apart from that the predictive abilities of the three scales in terms of enjoyment and impact are compared to each other to check which scale has superior predictive ability.

Participants. Sixty-one people participated in this experiment. Forty-five of them were female and 16 of them were male. Their ages ranged from 20 to 64 with a mean of 44.43 and a standard deviation of 13.95. They responded to calls for participants on social media websites (Facebook, LinkedIn) and on webpages dedicated to reading or books (Goodreads, Boeklezers.nl) or on personal pages of book clubs. These calls for participants asked for people who like to read and would like to participate in an experiment about reading experiences.

Design. The three scales that were needed to check the convergent validity of the SWAS were included in an experiment that investigated the effects of the use of suspense and curiosity discourse structures on story world absorption. This experiment used three conditions: a control condition and two experimental conditions in which the short story that was used was manipulated. Even though the factorial design of this experiment was developed with another study in mind, the data could still be used to check convergent validity, since the difference between the stories was not great.

Materials. A short story based on a fragment from the film *13Tzameti* was used in this experiment. This story is very suspenseful and therefore differs in style from the stories used in the first two studies. It has a total of 1157 words. The story was used in three different versions: one baseline version, one suspense version and one curiosity version.

Measures. The SWAS was administered, as well as the transportation scale (Green & Brock, 2000) and three dimensions of the narrative engagement scale (Busselle & Bilandzic, 2009) to check for convergent validity. All of the items were randomized on the questionnaire to prevent item order from influencing factor structure. The narrative understanding dimension of the narrative engagement scale was not included, because it does not have any reference to the SWAS. It is assumed in the present study, that when readers are absorbed they understand the story, since understandability is considered a precondition for story world

absorption to occur. The items used to measure enjoyment and impact in the first two studies were administered as well.

Procedure. The participants were sent an email that included a link to the online experiment on the site SurveyMonkey where the experiment was administered. They received instructions on how to fill out the questionnaires and were told to read in a relaxed manner as if they were reading something for their own pleasure. Then they were directed to the text they were asked to read. After they had read the story they filled out the questionnaires, which ended with the usual demographic questions (age, gender and education), correct reading questions (to check if they did really read the story), and a question about whether or not they knew the story.

Results

Convergent validity. The overall SWAS correlates highly with the narrative engagement scale ($r = .658$) as well as the transportation scale ($r = .685$). Among the sub-scales moderate to high correlations were found as well, ranging from .371 to .818 (all significant above $p < .01$). Table 9 contains correlations among the SWAS and its sub-scales, the narrative engagement scale and its sub-scales and the transportation scale. The strength of the correlations between the three scales demonstrates convergent validity of the story world absorption scale. It should be noted that the narrative understanding sub-scale of Busselle & Bilandzic's narrative engagement scale was not included in the present study, which can account for the overall high correlation between the three scales, whereas in earlier studies this sub-scale correlated only weakly or moderately with the transportation scale (Busselle & Bilandzic, 2009). All of the sub-scales of the SWAS, except emotional engagement, correlated highest with the narrative presence sub-scale of the narrative engagement scale, which might indicate a confounding of concepts in the narrative presence sub-scale of Busselle & Bilandzic. The emotional engagement sub-scale correlated highest with the transportation scale. The implications of these correlations are discussed further in the discussion section.

Comparing predictive abilities. In order to further assess the predictive validity of the SWAS its relations with enjoyment and impact were explored. Enjoyment correlated significantly with the overall SWAS ($r = .759$, $p = .000$), as did impact ($r = .781$, $p = .000$). To check whether the SWAS has better predictive power than other available scales, the same correlation analyses were conducted for the narrative engagement scale (Busselle & Bilandzic, 2009) and the transportation scale (Green & Brock, 2000). Enjoyment correlated significantly with the narrative engagement scale ($r = .327$, $p = .010$), as did impact ($r = .572$, $p = .000$). The transportation scale also correlated significantly with enjoyment ($r = .649$, $p = .000$) and

Table 9. Correlation analyses among all three scales and sub-scales in study 3

	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
1. Narrative Absorption full scale	1									
2. Attention sub-scale	.91**	1								
3. Emotional Engagement sub-scale	.92**	.78**	1							
4. Mental Imagery sub-scale	.76**	.61**	.66**	1						
5. Transportation sub-scale	.87**	.73**	.71**	.54**	1					
6. Narrative Engagement full scale (Without narrative understanding)	.66**	.53**	.65**	.50**	.60**	1				
7. Narrative Presence sub-scale	.81**	.82**	.65**	.64**	.71**	.66**	1			
8. Emotional Engagement sub-scale	.65**	.58**	.69**	.46**	.51**	.76**	.45**	1		
9. Attentional Focus sub-scale	.54**	.67**	.42**	.40**	.37**	-.03	.52**	.37**	1	
10. Transportation full scale	.69**	.51**	.70**	.66**	.55**	.54**	.41**	.59**	.24	1

impact ($r = .576$, $p = .000$). The correlations of all three scales with enjoyment were significant, but overall the SWAS correlated the highest. The results of the regression analyses can be found in Tables 10, 11 and 12 below.

Table 10. Results of regression analysis on enjoyment and impact for the story world absorption scale ($N=61$)

	Enjoyment	Impact
Story World Absorption	.76***	.78***
R²	.58	.61
F	80.35***	92.21***

* = $p < .05$, ** = $p < .01$, *** = $p < .00$

Table 11. Results of regression analysis on enjoyment and impact for the narrative engagement scale of Busselle & Bilandzic ($N=61$)

	Enjoyment	Impact
Narrative Engagement	.33*	.53***
R²	.11	.33
F	7.07*	28.72***

* = $p < .05$, ** = $p < .01$, *** = $p < .00$

Table 12. Results of regression analysis on enjoyment and impact for the transportation scale of Green & Brock (N=61)

	Enjoyment	Impact
Transportation scale	.65***	.58***
R²	.42	.33
F	43.01***	29.31***

* = $p < .05$, ** = $p < .01$, *** = $p < .00$

A comparison between two regression analyses on the sub-scales of the SWAS and the narrative engagement scale (the transportation scale lacks a clear dimensional structure) allows for further investigation of their relationship to enjoyment and impact. First enjoyment was regressed on the four sub-scales of the SWAS simultaneously and then the same was done for impact. The results of these analyses can be found in Table 13 below. The two analyses were repeated for Busselle & Bilandzic’s narrative engagement scale, the results of which can be found in Table 14 below. When comparing the two tables, we can see that even though the SWAS is better able to predict enjoyment and impact, there is still evidence of convergence. The emotional engagement sub-scale of both the SWAS and the narrative engagement scale does contribute to impact but not to enjoyment. Also the transportation sub-scale of the SWAS and the narrative presence sub-scale of the narrative engagement scale — which show conceptual overlap with each other — both contribute to enjoyment but not to impact. An interesting difference is that in the case of the narrative engagement scale only narrative presence contributes to enjoyment, whereas the regression analysis on the sub-scales of the SWAS shows a more nuanced picture, in which transportation, attention and mental imagery contribute to enjoyment.

Table 13. Results of regression analysis on enjoyment and impact for the sub-scales of the SWAS (N=61)

	Enjoyment	Impact
Transportation	.32*	-.09
Emotional engagement	.00	.61***
Mental imagery	.26*	.00
Attention	.31*	.33*
R²	.60	.68
F	20.92***	29.71***

* = $p < .05$, ** = $p < .01$, *** = $p < .00$

Table 14. Engagement scale (N=61) Results of regression analysis on enjoyment and impact for the sub-scales of the narrative

	Enjoyment	Impact
Narrative presence	.41**	.19
Emotional engagement	.09	.59***
Attentional focus	.21	.13
R²	.35	.58
F	10.22***	26.53***

* = $p < .05$, ** = $p < .01$, *** = $p < .00$

General discussion

This paper aimed to develop a measure that is specifically designed to measure the subjective experience of being absorbed in the story world of a narrative text during reading. The need for such an instrument arose from the need to further investigate the *nature* of the relationship between absorption experiences and enjoyment. Not only did we argue that it should be *empirically* established that enjoyment is indeed an outcome of (or response to) absorption rather than an aspect of such an experience, we also argued that the text a reader reads determines the particular evaluative response to a story world absorption experience. The instrument used to measure story world absorption, should therefore be able to predict different evaluative responses while at the same time being sensitive to various stimulus materials. This discussion section will first describe the SWAS and its four dimensions as identified by the first two studies. After that a summary and exploration of the various validity estimates used in this paper will be given to assess the convergent validity and predictive merits of the SWAS. We will end the discussion section by addressing the limitations of this study and deliver suggestions for further research.

The SWAS is a reliable instrument that can measure the same set of dimensions across different materials, while at the same time revealing the various ways in which it can differentiate between different absorbing experiences per narrative text that is read. The four aspects can be studied together as one experience or they can be investigated separately to reveal more information on particular texts and their effects. The first dimension is *attention* and describes readers' focus on the story world and consequent loss of awareness of the 'real world'. The second dimension labeled *emotional engagement* describes readers' emotional engagement with characters in the story. The third dimension, *mental imagery*, describes

whether readers have certain story-related imagery in mind while reading the story. The fourth dimension is *transportation* and describes the feeling of being transported to the story world.

Sensitivity to different materials: Relation between text, absorption and evaluation

An advantage of the SWAS over other scales is its division in four reliable sub-scales. This makes it possible to choose one's own course of investigation: combine the four aspects to investigate overall absorption or distinguish between them and only use one or more sub-scales. Granted, Busselle & Bilandzic's narrative engagement scale (2009) shares this advantage over other scales, however the present study demonstrates that the SWAS possesses greater predictive abilities and furthermore that the narrative engagement scale seems to suffer from dimensional confounding when it comes to its sub-scale of narrative presence. The narrative presence sub-scale combines items that in the SWAS belong to two different sub-scales: the transportation and the attention sub-scale. The dimensional structure of the SWAS also shows that we can speak of a separate dimension of transportation. Most studies on transportation use Green & Brock's definition of 'integrative melding of attention, imagery and feelings' (2000). However, the results of the present study show that not every reader who focuses their attention on a story or feels emotions in response to the story would also feel transported to that story world.

The results of the ANOVA's per story in each study show that the SWAS is sensitive to different materials and that it can be used to investigate the relationships between different texts and the type of story world absorption they inspire. For example, the action-oriented stories used in this paper elicited higher story world absorption scores overall than the character-oriented stories. When taking a closer look at the separate sub-scales, the difference between action-oriented stories and character-oriented stories is only significant in terms of transportation, attention and emotional engagement. It is interesting to notice that even though the overall scores were higher in the action-oriented stories and not in the character-oriented stories, emotional engagement is still the sub-scale with the highest scores. This, in our opinion, is an exciting direction for future research: what kinds of textual techniques are used to induce story world absorption and how do these techniques work? Perhaps action-oriented stories work better at eliciting story world absorption than character-oriented stories because they work to emotionally involve the reader, not in sympathizing with characters, but rather in following a suspenseful plot.

Apart from these results, the regression analyses conducted in this paper also tell us more about the dimensional structure of the SWAS and how this instrument could help the study of absorption's relationship to enjoyment and other outcomes. For example, the results of the present studies show that transportation does contribute to enjoyment but not to impact and that emotional engagement seems to be the overall strongest predictor of both enjoyment and impact. Future research that ventures to investigate this relationship between absorption and entertainment outcomes should try to broaden the scope of this field and include not only enjoyment, but other concepts that try to capture more complex evaluative responses in their studies. This paper can be seen as making an attempt at this by incorporating the concept of impact, however the impact concept needs further elaboration. For example, how does this notion relate to concepts such as appreciation and meaningfulness coined by Oliver & Bartsch (2010). Furthermore, how is the story world absorption scale related to these concepts? Can it be used to predict states of meaningfulness contemplation and appreciation? Our guess would be that stories that make use of foregrounding techniques would more easily elicit these types of states than stories that make use of more popular techniques such as suspense.

The present research endeavor already made a first step towards this type of research by using texts that make use of foregrounding techniques. The results show that those texts elicit lower story world absorption scores than texts that make use of popular techniques. However, since two different samples were used, it was not possible to compare these scores with an ANOVA. To be able to draw real conclusions with respect to this hypothesis, future studies in this direction should first look into manipulating degrees of foregrounding in a text.² The regression analyses showed that story world absorption was a better predictor of both enjoyment and impact when used in combination with more foregrounded stories. Of course this does not tell us anything about a causal relation, but it does inspire reflection on the nature of the relationships between text, absorption and entertainment outcome. Our suggestion for further research in this direction would be to use more advanced methods, such as path analyses to find out more about the causal relationships within this network of related concepts.

Another interesting avenue for further research would be to investigate the SWAS' ability to predict after effects such as persuasion. This was not the focus of the present study, but could nevertheless be a promising addition to research on relationships between texts, absorption and evaluative responses. If a certain text is enjoyed more than the other is it than also more succesful in persuading the reader of a certain message? Are there fictional texts that are less succesful in persuasion than others, because they do not lead to enjoyment, but to impact? These are just a few examples of questions that come to mind.

Convergent validity

Since the construction of the SWAS largely builds on the foundations of similar research done by Green & Brock and Busselle & Bilandzic, it was expected to correlate with these scales and show convergent validity. Correlation analyses indeed showed convergent validity of the SWAS since it correlated strongly with the transportation scale (Green & Brock, 2000) and the narrative engagement scale (Busselle & Bilandzic, 2009). Future research should first look into the discriminant validity of the SWAS. Now that we know that it converges with scales that try to capture a similar concept, it is important to learn whether the scale is divergent enough from instruments that try to measure contrasting concepts.

Predictive validity

Even though the correlations between the SWAS and the transportation and narrative engagement scales were rather high, there is still a divergent factor distinguishing the SWAS from the other two scales, namely the SWAS' ability to predict different entertainment outcomes. The third study presented in this paper showed that the SWAS was *more* successful in predicting both enjoyment and impact than Busselle & Bilandzic's narrative engagement scale and Green & Brock's transportation scale. What is interesting is that the regression analyses conducted on the sub-scales of the SWAS and the sub-scales of the narrative engagement scale, still showed convergent validity. For example, both the emotional engagement sub-scale of the narrative engagement scale and of the SWAS predicted impact, but not enjoyment. However, overall the SWAS was able to predict 76 percent of variance in enjoyment and 78 percent of variance in impact, whereas the narrative engagement scale was only able to predict 33 percent of variance in enjoyment and 57 percent of variance in impact. The transportation scale was able to predict 65 percent of variance in enjoyment and 58 percent of variance in impact. In other words, there seems to be more conceptual similarities between the SWAS and the transportation scale than between the SWAS and the narrative engagement scale, which could be due to the inclusion of a mental imagery sub-scale. Overall, the SWAS has the strongest predictive abilities out of the three measures that were tested.

Limitations and outlook

Even though, the diversity in participants and materials used gives confidence in the usability of the scale in various samples and with different types of narrative texts, the scale could be tested with a broader set of samples beyond high school

students and University students from the Netherlands. Also, this instrument could be combined with various types of stories in future research. One suggestion for further research would be to test the scale in combination with longer stories. It is very plausible that longer stories result in increased absorption.

Another limitation of this scale is that it can — in its current form — only be used in combination with *written* narratives. A simple solution to this problem would be to adapt the story world absorption scale to the demands of other narrative media by simply eliminating or adding certain dimensions. For example, adding the mental imagery dimension when testing written texts and leaving it out when testing films could be a viable option. The topic of adapting scales for use with different narrative media has been under-appreciated in the relevant literature so far. For example, the narrative engagement scale is developed with a *cinematic* medium in mind and therefore Busselle & Bilandzic do not include aspects that are necessary for absorption in a *textual* medium (2009). Green & Brock do include a mental imagery sub-scale, because in their earlier studies they only measure narrative experiences with *textual* media (2000). In later studies (Green, Brock & Kaufman, 2004; Green, Kass, Carrey, Herzig, Feeney & Sabini, 2008) the difference in media is addressed and empirically tested. In their 2008 study on repeated exposure to print and film, Green and her colleagues point out that ‘transportation includes mental imagery but it remains an open question whether it is important for that imagery to be self-generated (as in print) or provided by the medium (as in film)’ (Green et al., 2008). The nature of the two media — literature and film — and especially the demands that they make of their recipients are essentially different and therefore a questionnaire asking about a narrative *viewing* experience would have to be different in essence from a questionnaire asking about a narrative *reading* experience.

These limitations notwithstanding, this research presents the groundwork for the development and use of a new measure that can contribute to the study of story world absorption experiences. An exciting new avenue for further research in our opinion is the possibility of using this instrument to investigate the specific relationships between a certain text and its subsequent absorbing experience and evaluative response. This paper can be seen as a first step in that direction; however, in this study existing stories were used without manipulating specific textual features. This prevented a straight comparison between specific story elements. In future studies it would be appropriate to work with specific narrative techniques, manipulating them to pinpoint exactly which techniques lead to which kind of configuration of story world absorption and what kind of subsequent evaluative response.³ Apart from that it would be interesting to further investigate the relationships of the four aspects of story world absorption to each other. The correlations between the four dimensions on the SWAS are only modest, which indicates

that the dimensions are related, but not similar to each other. The question remains whether the four dimensions are causally related to each other in ways that have yet to be determined. For example: does transportation follow attention or does emotional engagement lead to mental imagery and vice versa? Furthermore, the divergent results of the regression analyses for each of the four stories seem to indicate that the causal relations between the four dimensions might differ from one story to the next. In sum, the story world absorption scale is a reliable instrument that offers great possibilities to explore various avenues of research into absorption experiences, their textual determinants and entertainment outcomes.

Acknowledgements

The authors are grateful to Keith Oatley and six anonymous reviewers for their comments on earlier drafts of this manuscript.

Notes

1. There is a large literature on different modalities of imagery that could possibly play a role in absorbing reading experience, mainly motor imagery (Zwaan, 2003; Kuijpers & Miall, 2011). However, since the focus of the present study is on mental imagery it was chosen not to include this issue in the present paper because it stretches beyond the confines of the theoretical framework.
2. A forthcoming paper by Kuijpers et al (*Effects of textual deviation on story world absorption*) deals with the question of how foregrounding influences story world absorption by manipulating degrees of deviation in several stories.
3. A forthcoming paper by Kuijpers & Doicaru (*Suspense and curiosity in narrative film and literature: a media comparison*) investigates the effects of manipulated suspense or curiosity discourse structures on aspects of story world absorption.

References

- Balakrishnan, B., & Sundar, S. S. (2011). Where am I? How can I get there? Impact of navigability and narrative transportation on spatial presence. *Human-Computer Interaction*, 26(3), 161–204.
- Bilandzic, H., & Busselle, R. W. (2008). Transportation and transportability in the cultivation of genre-consistent attitudes and estimates. *Journal of Communication*, 58(3), 508–529. DOI:10.1111/j.1468-2885.2008.00322.x
- Brewer, W.F., & Lichtenstein, E.H. (1982). Stories are to entertain: A structural-affect theory of stories. *Journal of Pragmatics*, 6 (5-6), 473–486. DOI: 10.1016/0378-2166(82)90021-2

- Busselle, R., & Bilandzic, H. (2009). Measuring narrative engagement. *Media Psychology*, 12(4), 321–347. DOI:10.1080/15213260903287259
- Csikszentmihalyi, M. (1988). *The flow experience and its significance for human psychology*, Cambridge University Press.
- Cohen, J. (2001). Defining identification: A theoretical look at the identification of audiences with media characters. *Mass Communication and Society*, 4(3), 245–264. DOI: 10.1207/S15327825MCS0403_01
- Duchan, J. F., Bruder, G. A., & Hewitt, L. E. (1995). *Deixis in narrative: A cognitive science perspective*, Lawrence Erlbaum Associates Hillsdale.
- Field, A. (2005) *Discovering Statistics using SPSS. Second Edition*. Sage Publications, London.
- Gerrig, R. J. (1993). *Experiencing narrative worlds: On the psychological activities of reading*. Yale University Press.
- Green, M. C., & Brock, T. C. (2000). The role of transportation in the persuasiveness of public narratives. *Journal of Personality and Social Psychology*, 79(5), 701–721. DOI: 10.1037//0022-3514.79.5.701
- Green, M.C., Brock, T.C. & Kaufman, G.F. (2004) Understanding media enjoyment: The role of transportation into narrative worlds. *Communication Theory*, 4, 311–327. DOI: 10.1111/j.1468-2885.2004.tb00317.x
- Green, M. C., Kass, S., Carrey, J., Herzig, B., Feeney, R., & Sabini, J. (2008). Transportation across media: Repeated exposure to print and film. *Media Psychology*, 11(4), 512–539. DOI: 10.1080/15213260802492000
- Hakemulder, J. F. (2004). Foregrounding and its effect on readers. *Discourse Processes*, 38(2), 26.
- Hall, A. E., & Bracken, C. C. (2011). “I really liked that movie”: Testing the relationship between trait empathy, transportation, perceived realism, and movie enjoyment. *Journal of Media Psychology: Theories, Methods, and Applications*, 23(2), 90.
- Hall, A., & Zwarun, L. (2012). Challenging entertainment: Enjoyment, transportation, and need for cognition in relation to fictional films viewed online. *Mass Communication and Society*, 15(3), 384–406. DOI: 10.1080/15205436.2011.583544
- Knobloch, S., Patzig, G., Mende, A. M., & Hastall, M. (2004). Affective news effects of discourse structure in narratives on suspense, curiosity, and enjoyment while reading news and novels. *Communication Research*, 31(3), 259–287. DOI: 10.1177/0093650203261517
- Kuijpers, M.M. & Miall, D.S. (2011). Bodily Involvement in Literary Reading: An Experimental Study of Readers’ Bodily Experiences during Reading. In F. Hakemulder (Eds.), *De Stralende Lezer. Wetenschappelijk Onderzoek naar de Invloed van het Lezen* (pp. 160–174). Delft: Eburon, IGEL.
- Lee, K. M. (2004). Presence, explicated. *Communication Theory*, 14(1), 27–50. DOI: 10.1111/j.1468-2885.2004.tb00302.x
- Miall, D. S., & Kuiken, D. (1995). Aspects of literary response: A new questionnaire. *Research in the Teaching of English*, 29(1), 37–58.
- Nabi, R. L., & Krcmar, M. (2004). Conceptualizing media enjoyment as attitude: Implications for mass media effects research. *Communication Theory*, 14(4), 288–310. DOI: 10.1111/j.1468-2885.2004.tb00316.x
- Nell, V. (1988). *Lost in a book: The psychology of reading for pleasure*. New Haven, CT: Yale University Press.
- Oliver, M.B. & Bartsch, A. (2010) Appreciation as audience response: Exploring entertainment gratifications beyond hedonism. *Human Communication Research*, 36, 53–81. DOI: 10.1111/j.1468-2958.2009.01368.x

- Ravaja, N. (2004). Contributions of Psychophysiology to Media Research: Review and Recommendations. *Media Psychology*, 6, 193–235. DOI: 10.1207/s1532785xmep0602_4
- Segal, E. M. (1995). *Narrative comprehension and the role of deictic shift theory*. Deixis in Narrative: A Cognitive Science Perspective, 3–17.
- Tal-Or, N., & Cohen, J. (2010). Understanding audience involvement: Conceptualizing and manipulating identification and transportation. *Poetics*, 38(4), 402–418. DOI: 10.1016/j.poetic.2010.05.004
- Tamborini, R., Bowman, N. D., Eden, A., Grizzard, M., & Organ, A. (2010). Defining media enjoyment as the satisfaction of intrinsic needs. *Journal of Communication*, 60(4), 758–777. DOI: 10.1111/j.1460-2466.2010.01513.x
- Tan, E. S., & Fasting, B. T. (1996). *Emotion and the structure of narrative film: Film as an emotion machine*. Lawrence Erlbaum Associates, Inc.
- Tellegen, A., & Atkinson, G. (1974). Openness to absorbing and self-altering experiences (“absorption”), a trait related to hypnotic susceptibility. *Journal of Abnormal Psychology*, 83(3), 268. DOI: 10.1037/h0036681
- Vorderer, P., Wulff, H. J., & Friedrichsen, M. (1996). *Suspense: Conceptualizations, theoretical analyses, and empirical explorations* Lawrence Erlbaum.
- Wolf, W. (2004). Aesthetic illusion as an effect of fiction. *Style-Fayetteville*, 38(3), 325.
- Zwaan, R. A. (2003). The immersed experiencer: Toward an embodied theory of language comprehension. *Psychology of Learning and Motivation*, 44, 35–62. DOI: 10.1016/S0079-7421(03)44002-4

Corresponding author's address

Moniek M. Kuijpers
Muntstraat 2A
3512 EV
Utrecht
The Netherlands
M.M.Kuijpers@uu.nl